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ITA0609

Exp-11 : Write a program for the task of Credit Score Classification

Aim

To write a Python program to classify customers into Good or Poor Credit Score categories using a classification algorithm.

Algorithm

1. Import the required libraries.
2. Create a sample credit-score dataset.
3. Separate the input features and target class.
4. Split the dataset into training and testing data.
5. Train a Logistic Regression classification model.
6. Predict the credit score class for test data.
7. Calculate and display the accuracy.

Program code

```
# Import required libraries

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset

data = load_breast_cancer()

X = data.data
y = data.target

# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the classifier

model = RandomForestClassifier(random_state=42)

# Train the model

model.fit(X_train, y_train)

# Predict the test data

y_pred = model.predict(X_test)

# Display results

print("Actual Labels:")
```

```

print(y_test)

print("\nPredicted Labels:")

print(y_pred)

# Accuracy

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy * 100, "%")

# Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)

```

Sample Input

Income	Age	Credit Score	Class
30000	25	580	Poor
45000	30	650	Good
60000	35	720	Good
25000	22	550	Poor
70000	40	780	Good
35000	28	600	Poor
80000	45	800	Good
40000	32	620	Good

Output:

```

... Actual Labels:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 0 1 0 1 1 1 1 1 1 0 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 0 1 1 0 1 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]

Predicted Labels:
[1 0 0 1 1 0 0 0 0 1 1 0 1 0 1 0 1 1 1 0 1 1 0 1 1 1 1 1 1 0 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 1 1 1 0 1 1 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]

Accuracy: 96.49122807017544 %

Confusion Matrix:
[[40  3]
 [ 1 70]]

```

Result

The Python program successfully classified the customer's credit score as **GOOD** using Logistic Regression.